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Workflow Automation Properties

When creating workflow automations (Automation Rules and SLAs), users need to configure:

- **Input** (Automation Rules) or **Time and Schedule** (SLAs) to define if the automation will trigger based on a specific action or time, respectively.
- **Conditions** (Automation Rules) or **Breach Conditions** (SLAs) to define the event or case fields to be considered in the automation.
- **Outputs** (Automation Rules) or **Breach Actions** (SLAs) to define what should happen when the automation is triggered.

This page details the available configuration options.

Inputs (Automation Rules)🔗🔗

This section lists the available automation rule inputs for events and cases.

Events🔗🔗

Input	Description
Event Access Group Changes	When the access group ID of an event is changed.
Event Arrives	When a new event arrives from Pulse.
Event Assignee Changes	When an event is assigned to an analyst.
Event Note is Added	When a note is added to an event.
Event Queue Changes	When an event is moved into a queue.
Event Status changes	When the status of an event is changed.
Event Updated	When the fields of an event are updated.
Most Recent Event Version Received	When the received version of the event is a new event for the system or it is more recent than the existing on the system. This input works similar to "Event Updated" but it ignores the out-of-order event updates. Learn more on how Case Manager's processes events .

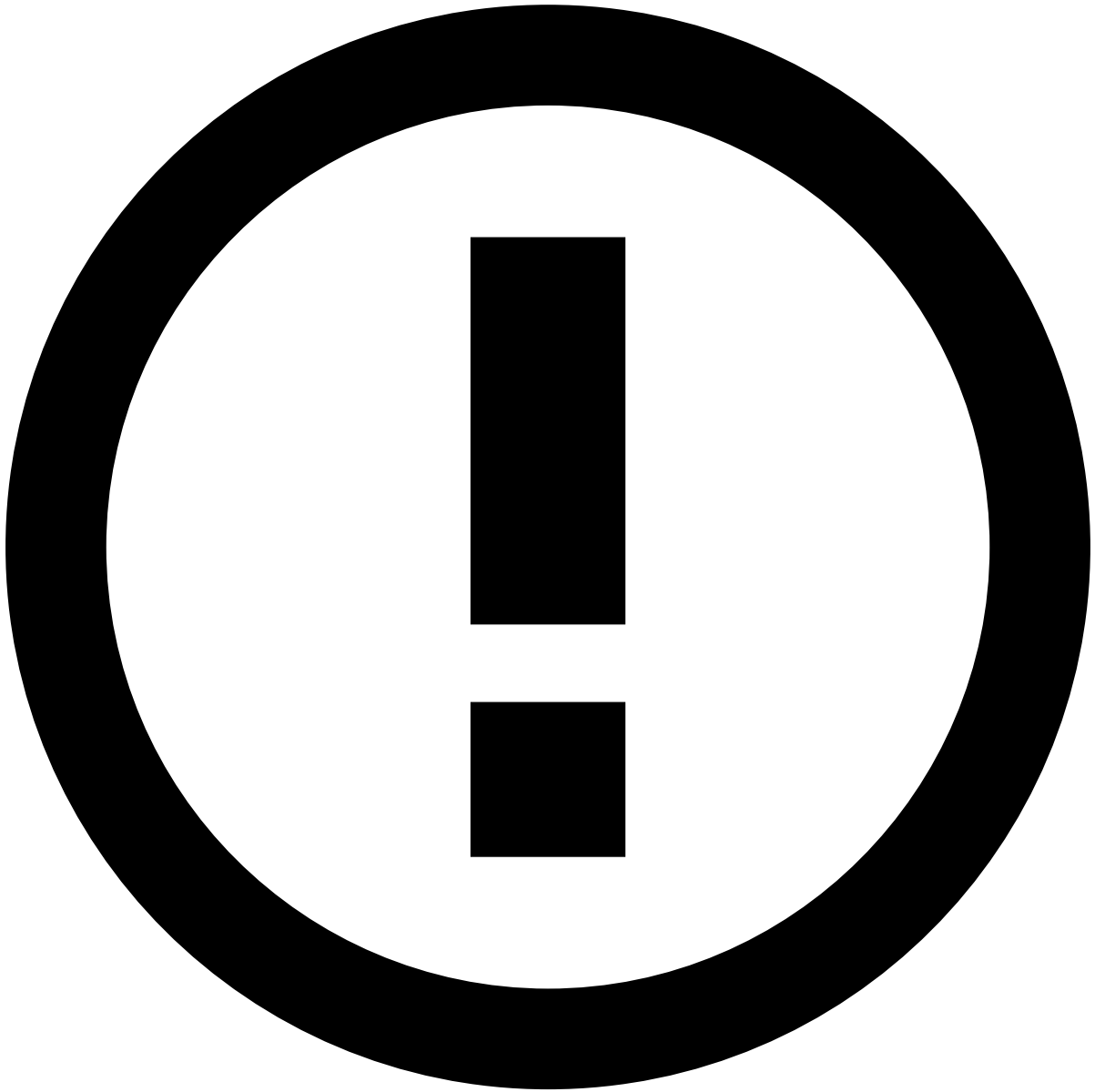
Cases🔗🔗

Input	Description
Case Assignee Changes	When a case is assigned to an analyst.
Case Created	When a case is created.
Case Note is Added	When a note is added to a case.
Case Queue Changes	When a case is moved into a queue.
Case Status Changes	When the status of a case is changed.
Case Updated	When the fields of a case are updated.

Time and Schedule (SLAs)📄

This section details when an SLA can be triggered. Note that, events are only processed by an SLA when the system includes an Automation Rule that is configured to start that SLA.

Input	Description
Time Period	The SLA is triggered after a certain period since the corresponding automation rule was triggered: <i>Days, Hours or Minutes</i>. Imagine that you want to send all incoming alerts that are not reviewed within 24 hours to a specific queue. To do this, configure an SLA with a 24 hours Time Period and then configure an automation rule to mark the alerts to be evaluated by the SLA, using the "Start SLA" output. This means that, when an alert arrives to Case Manager, the automation rule will trigger and start the 24 hour period. After that time, if the alert has not been reviewed, it will automatically be sent to the defined queue.
Time Schedule	The SLA is triggered recurrently, following the defined frequency since the corresponding automation rule was triggered: <i>Daily, Daily except weekends or Weekly</i>. Imagine that you want to send all incoming alerts to a specific queue, every day at 8 am. To do this, configure an SLA with a daily Time Schedule and then configure an automation rule to mark the alerts to be evaluated by the SLA, using the "Start SLA" output. This means that, when an alert arrives to Case Manager, the automation rule will trigger and, if the alert has not been reviewed until the defined scheduled, it will automatically be sent to the defined queue.



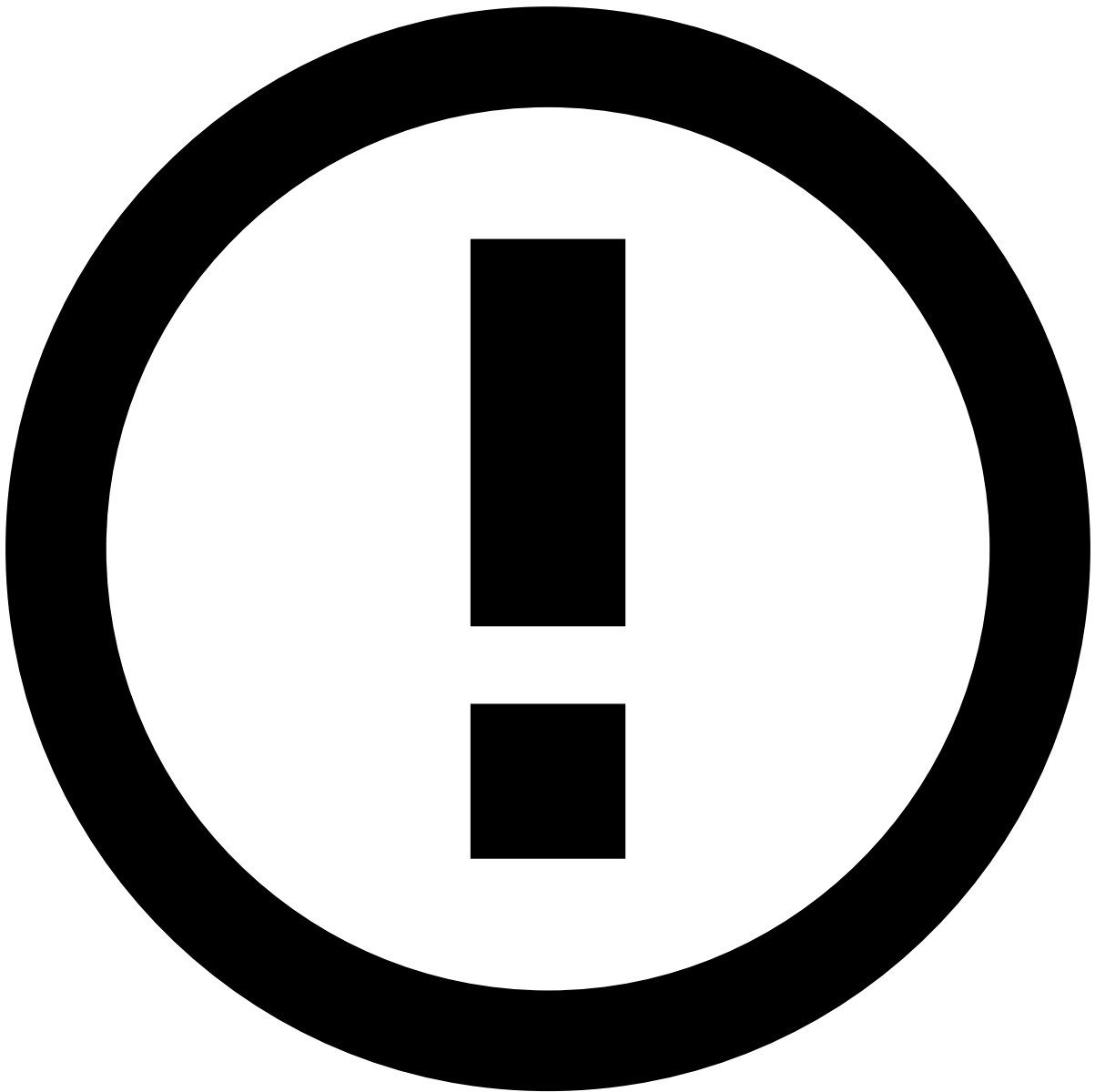
Exclude weekends

When this option is selected, the SLA does not take weekends (Saturdays and Sundays) into consideration. For instance, a 24-hour SLA due on a Saturday at 2:00 PM will be triggered on the following Monday at 2:00 PM. Likewise, a 48-hour SLA due on a Sunday will be triggered on the following Tuesday.

Conditions (Automation Rules and SLAs)

The conditions assess the fields of the event or case to identify if the actions should be executed or not. Each condition includes the following components:

- **Event or Case field** to be used as target value to be compared with the *expected value*.
- **Operator** to assert the expected value against the target value. See *Available Operators* in the [Manage Automation Rules](#) and [Manage SLAs](#) pages for more information.
- **Expected Value** to be compared with the event field's value.



Hiding fields in automation rules conditions

It's possible to control which event fields are visible in automation rules conditions. This is typically needed for fields with encrypted data but it may also be beneficial to hide fields unavailable to end users. Learn more on [how to hide fields in automation rules conditions](#).

Actions (Automation Rules and SLAs)🔗📄

This section lists the available actions for events and cases.

Events🔗📄

Action	Description
Add to list	Add an event's field value to a Block/Allow List . Items can have an <code>expiration period</code> to specify how many days an item should be held on the list. This is an optional configuration and, if no expiration period is provided, no expiration date will be set for the item.

Action	Description
Call Webhook	Calls an external system by making an HTTP POST request to the provided URI. Note that within the request's payload is the event that triggered the rule (JSON format).
Create Case or link to existing Case	Links the event to a new or existing case.
Execute Extension	Executes a custom extension according to Case Manager's configuration. For example, send an SMS to the client.
Link to Existing Case	Links the event to an existing case.
Send to Queue	Send the event into a given queue.
Set Access Group	Change the access group ID of an event.
Set Event Assignee	Assign the event to an analyst.
Set Initial Status	Set the initial status of an event and, optionally, a set of reasons associated to the status change. Additionally, it notifies Pulse about that status change. Note that, depending on your Case Manager configuration, this list may include other actions related to the event status. For example, <i>Set Initial Acceptance Status</i> or <i>Set Initial Review Status</i> .
Set Status	Set the status of an event and, optionally, a set of reasons associated to the status change. Additionally, it notifies Pulse about that status change. Note that, depending on your Case Manager configuration, this list may include other actions related to the event status. For example, <i>Set Acceptance Status</i> or <i>Set Review Status</i> .
Start SLA	Starts a new SLA automation.

Cases

Action	Description
Call Webhook	Calls an external system by making a HTTP POST request to the provided URI. Note that within the request's payload is the event that triggered the rule (JSON format)
Execute Extension	Executes a custom extension according to Case Manager's configuration. For example, send an SMS to the client.
Send to Queue	Send the case into a given queue.
Set Case Assignee	Assign the case to an analyst.
Set Status	Set the status of a case. Additionally, it notifies Pulse about that status change.
Set Status of all linked events	Set the status of all events linked to a case and, optionally, a set of reasons associated to the status changes. Additionally, it notifies Pulse about those status changes.
Start SLA	Starts a new SLA automation.

The automation rules' actions are executed over an event snapshot taken when the action was triggered. This means that the actions of multiple automation rules executed during a single trigger (e.g. event status changed) will not view the modifications performed by each other - only a subsequent trigger will have an updated view of the event.

Imagine automation rules A and B that are both defined to trigger on event arrival. According to the configuration, Automation A should send the event to queue A while Automation B should send the event to queue B. According to the defined order, automation A should execute before automation B, assuming the execution of automation rules in parallel is disabled (see below).

When an event arrives in CM, these automations will be triggered. Considering that both their conditions are satisfied, the event is sent to queue B (since automation B was the last to run).

However, note that the activity log of that event will not display a queue transition from "none" to "queue A" and then to "queue B" because both actions will be executed over the snapshot of the event as it was when it triggered the automation rules. This means that, when the event arrived, its queue was "none", which is why this behavior occurs. The same will be

observed in the database (AM_EVENT_QUEUE), which will show two separate queue changes (to queue A and to queue B) where the OLD_QUEUE column will be null.

To summarize, different actions belonging to the same trigger don't depend on each other.

Note that this behavior only happens PER trigger, but not across multiple automation rule triggers.



Automation rules execution in parallel

In order to increase automation rules processing throughput, it's possible to set up CM to execute multiple automation rules at a time (ie. to enable ARs to be processed in parallel). Allowing automation rules to be processed in parallel allows to increase the number of rules processed per second. However, such parallelism will compromise CM's ability to ensure the automation rules are always executed according their defined order. This means that, when parallel execution of automation rules is in place, rules triggered nearly the same time (within some seconds apart of each other) may or may not be executed at the pre-defined order (although all rules are always executed).

This trade-off between increasing the automation rules throughput and ensuring a deterministic execution order of the rules is configured at config.json file, through the following parameters

- `performance.automation_worker_threads` (default value is 4)
- `performance.automation_actions_batches.batch_threads` (default value is 1)

Setting both these parameters to `1` will ensure automation rules will be executed by the defined order. On the other hand, increasing these values allows to process more rules in parallel (foregoing the rules to be executed by a pre-defined order).

Check [this page](#) for more details on the above config.json parameters.

Learn More📖

Learn more on how to configure [Automation Rules](#) and [SLAs](#).